



Chiral Gauge Theory on the Lattice

Vatsalya Vaibhav
with Prof. Latham Boyle
Higgs Centre for Theoretical Physics

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- Fermion Doubling
- Nielsen Ninomiya theorem
- Sign problem

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- Anomaly structure

○ Worldline KD fermions and the sign problem

Fermion Doubling

- Naive discretization of fermions

$$S = a^4 \sum_n \bar{\Psi}_n \left(\sum_{\mu=1}^4 \gamma_{\mu} \frac{\Psi_{n+\mu} - \Psi_{n-\mu}}{2a} + m \Psi_n \right)$$

- Gives the propagator

$$\left(\sum_{\mu=1}^4 \gamma^{\mu} \frac{1}{a} \sin(k_{\mu} a) \right)^{-1}$$

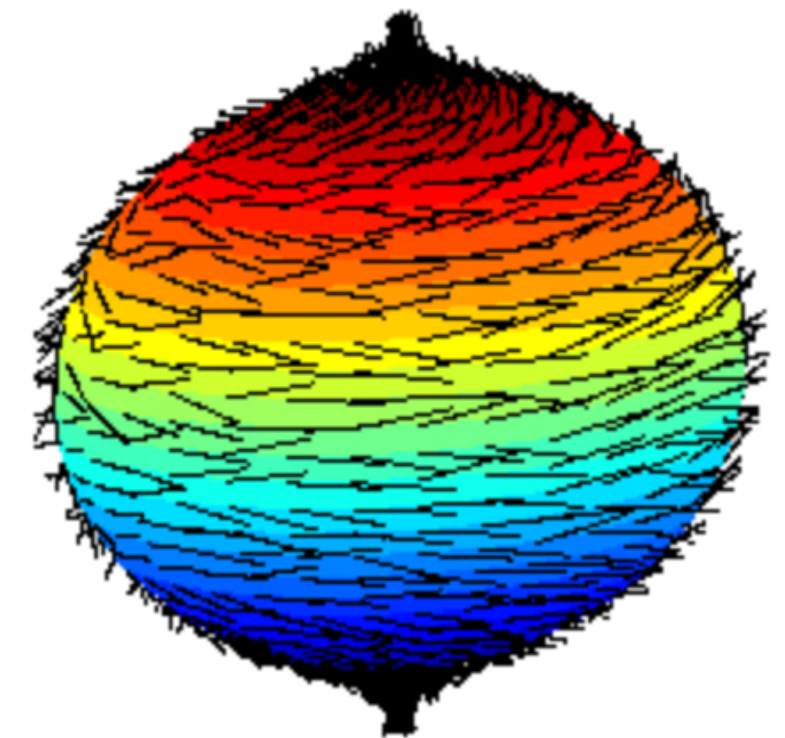
- Gives 16 poles per Brillouin zone resulting in 16 extra fermions than in the continuum theory

Nielsen-Ninomiya Theorem

Theorem: Under the assumption of locality, hermiticity, and periodicity, any lattice description of chiral fermions necessarily leads to equal number of left and right-handed fermions.

- Euclidean proof (Poincaré-Hopf theorem)

$$\sum \underbrace{\text{ind}(iD)}_{\substack{\downarrow \\ \text{continuous vector field}}} = n_L - n_R = \chi \substack{\downarrow \\ \text{Euler characteristic}} = 0 \text{ for a torus}$$



Sign Problem

- Lorentzian spacetime

$$\begin{aligned} \text{Spin}(3,1) &\cong \text{sl}(2, \mathbb{C}) \\ &\cong \text{su}(2) \otimes \mathbb{C} \end{aligned}$$

- $S_L = \int d^4x i \bar{\Psi}_L \sigma^\mu \partial_\mu \Psi_L$
- $\Psi_L \rightarrow U \Psi_L, \bar{\Psi}_L^\dagger \rightarrow \bar{\Psi}_L^\dagger U^\dagger$
 $\sigma^\mu \partial_\mu \rightarrow U \sigma^\mu \partial_\mu U^\dagger$

- Euclidean space

$$\text{Spin}(4) \cong \text{SU}(2) \times \text{SU}(2)$$

- $S_E = \int d^4x i \bar{\chi}_R \sigma^\mu \partial_\mu \Psi_L$
- $\Psi_L \rightarrow U \Psi_L, \bar{\chi}_R \rightarrow \bar{\chi}_R V^\dagger$
 $\sigma^\mu \partial_\mu \rightarrow V \sigma^\mu \partial_\mu U^\dagger$

Sign Problem

- Lorentzian spacetime

$$\begin{aligned} \bullet \quad Z_L &= \int D\Psi_L^\dagger D\Psi_L e^{iS_L} \\ &= \det(\sigma^\mu \partial_\mu) \\ &\quad \downarrow \\ &\det(u \sigma^\mu \partial_\mu u^\dagger) \end{aligned}$$

- Euclidean space

$$\begin{aligned} \bullet \quad Z_E &= \int D\chi_R D\Psi_L e^{-S_E} \\ &= \det(\sigma^\mu \partial_\mu) \\ &\quad \downarrow \\ &\det(v \sigma^\mu \partial_\mu v^\dagger) \\ &= e^{i\phi} \det(\sigma^\mu \partial_\mu) \end{aligned}$$

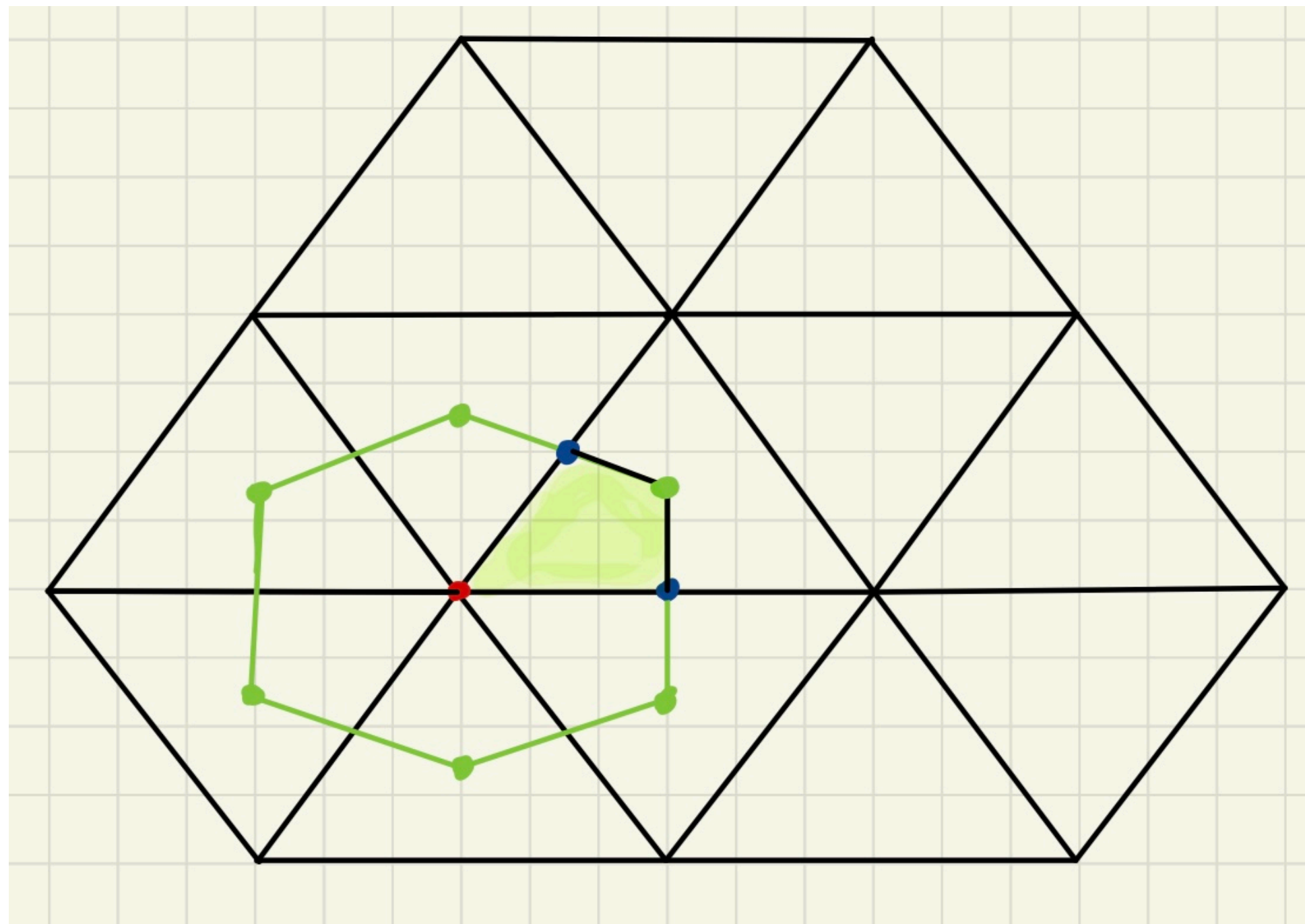
$$Z_E = e^{i\phi} |Z_E| = e^{i\phi} Z_L$$

Motivation

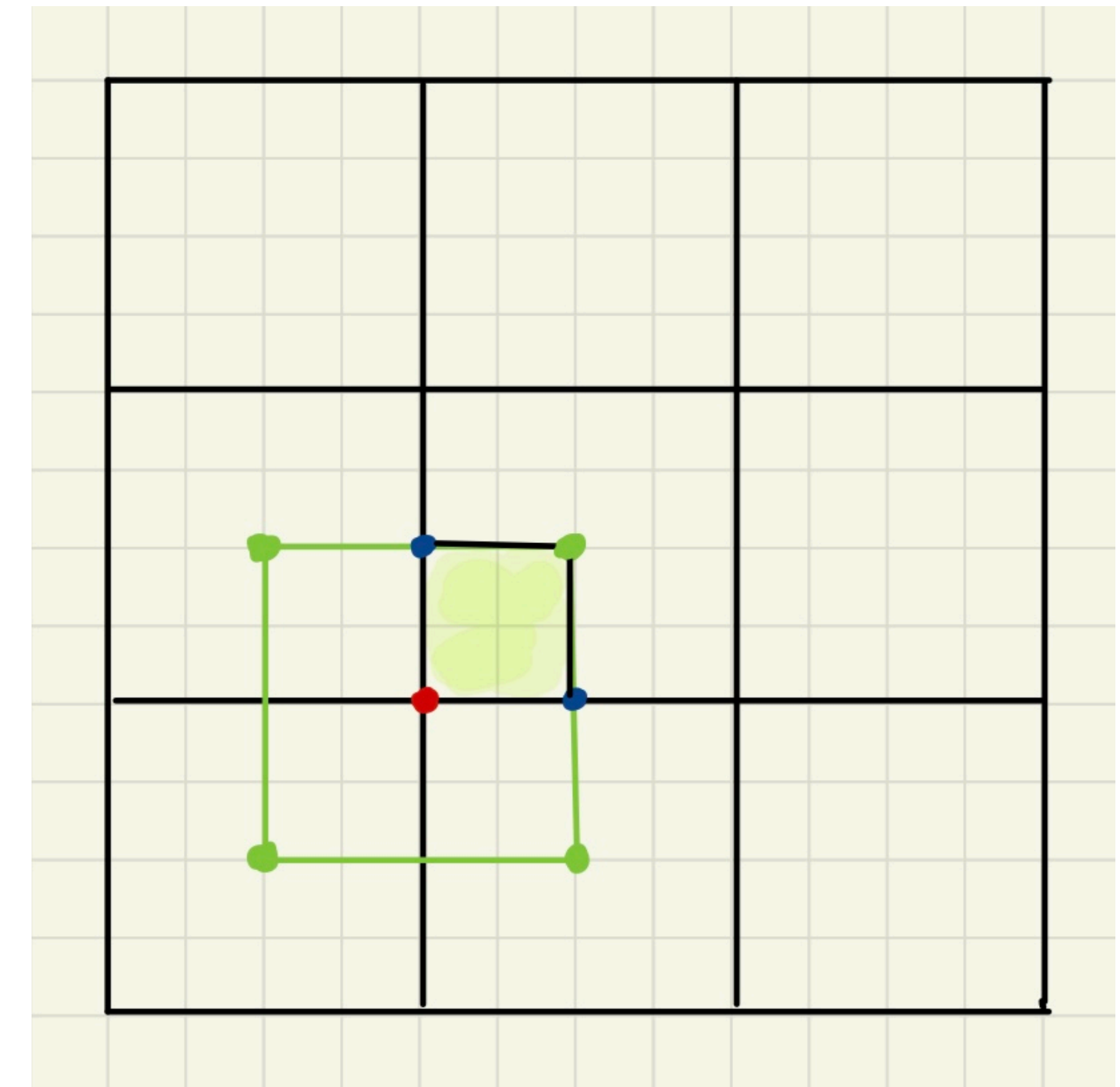
- All these problems seem to suggest a serious lack of understanding in the fundamental geometrical aspects of fermions and their realisation on the lattice.
- These problems do not appear in the discretization of bosons in lattice field theory. The following strategy seems to work well in the case of bosons:
 1. Place 0-form scalars on the vertices (0-sites)
 2. Place the 1-form vector fields (like gauge bosons) on the edges (1-sites)
 3. Place the 2-form tensor fields (like the field-strength tensor) on the plaquets (2-sites)
- Where should fermions be placed? Kahler-Dirac fermions provide a nice geometrical discretization of fermions as polyform fields. (See T. Banks 1982, J. Rabin 1982, Becher and Joos 1982)

Geometric discretisation of Polyforms

$$\phi = \phi + \phi_\mu dx^\mu + \phi_{\mu\nu} dx^\mu \wedge dx^\nu$$



(3,6) triangular lattice



(4,4) square lattice

Preserved (Co)Homology on the Lattice

- Incidence function

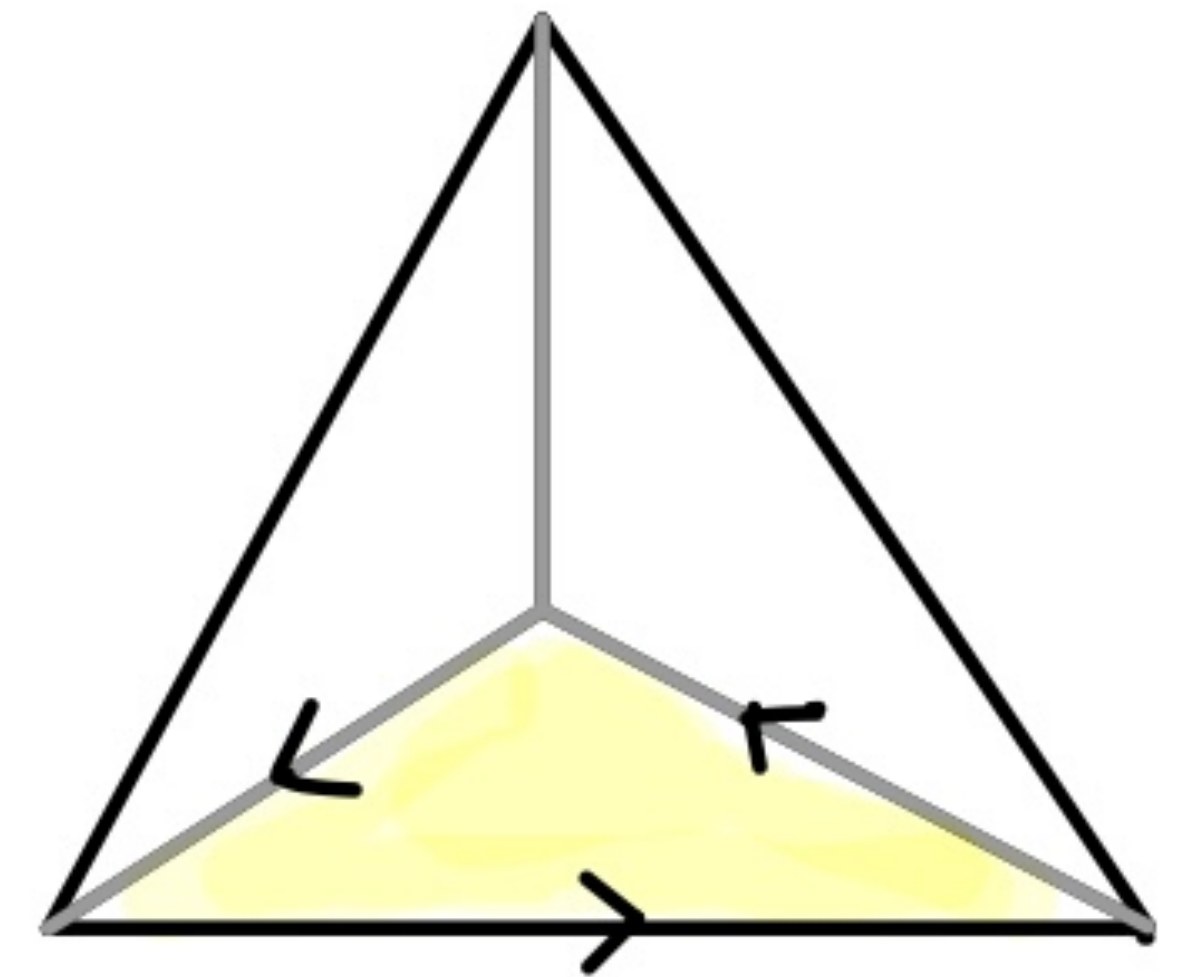
$$I(C_{p-1}, C_p) = \begin{cases} +1 \\ -1 \\ 0 \end{cases}$$

- The Hodge star

$$*I(*C_{p-1}, *C_p) = I(C_{D-p}, C_{D-p+1})$$

- Cochains

$$\Phi: C_p \rightarrow \mathbb{C} ; \quad *\Phi(*C_p) = \Phi(C_{D-p})$$



A 2-simplex and its boundary

Preserved (Co)Homology on the Lattice

- Exterior derivative

$$d\Phi(C_p) = \sum_{C_{p-1}} I(C_{p-1}, C_p) \Phi(C_{p-1})$$

- Co-exterior derivative

$$*d*\Phi(C_p) = \sum_{C_{p+1}} I(C_p, C_{p+1}) \Phi(C_{p+1})$$

$$d^2 = 0 \ ; \ (*d*)^2 = 0$$

- Wedge product

$$\Phi \wedge \Psi(C_{n+m}) = \frac{1}{(n+m)!} \sum_{\sigma \in \text{perm}(n+m)} \text{sgn}(\sigma) \Phi(C_{\sigma(n)}) \Psi(C_{\sigma(m)})$$

Lattice Action

- Kahler-Dirac operator

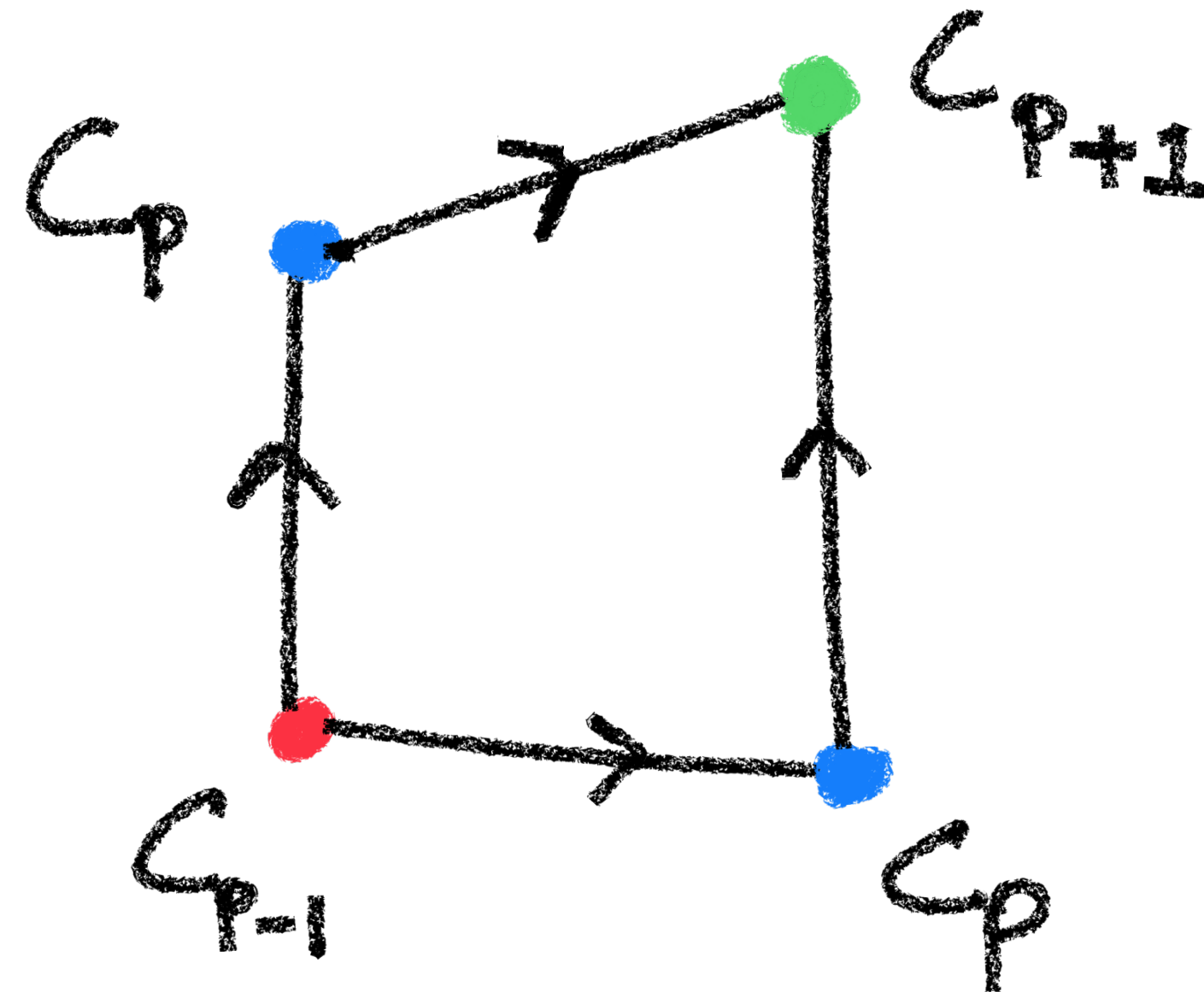
$$K = (d - *d*)$$

- Continuum action

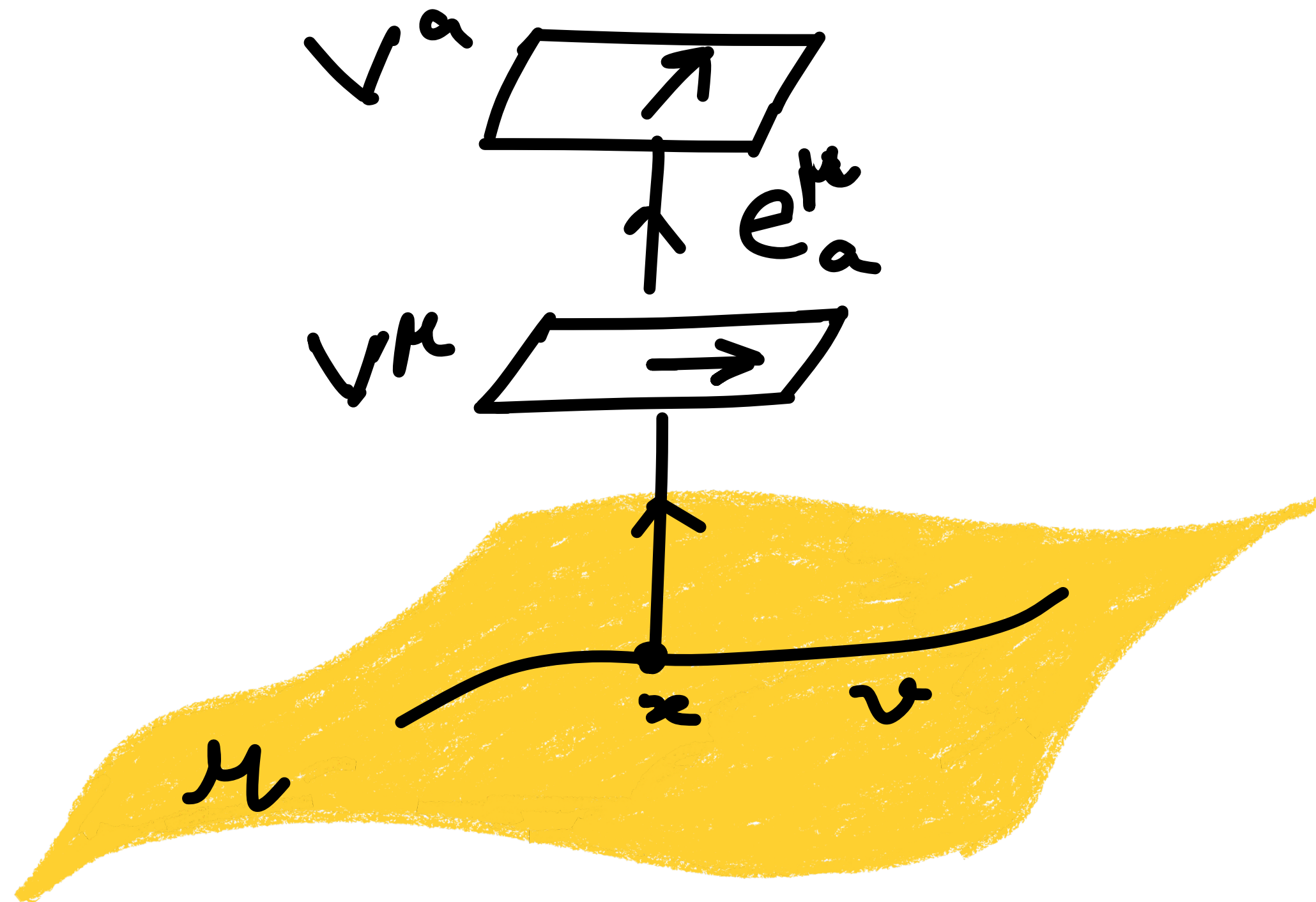
$$S = \int_M \bar{\Phi} \wedge (d - *d*) \Phi$$

- Lattice action

$$S = \sum_p \sum_{\zeta_p} \bar{\Phi}(\zeta_p) \left[\sum_{\zeta_{p-1}} I(\zeta_{p-1}, \zeta_p) \Phi(\zeta_{p-1}) + \sum_{\zeta_{p+1}} I(\zeta_p, \zeta_{p+1}) \Phi(\zeta_{p+1}) \right]$$



The Local Group of Rotations



$$e_a^\mu \longrightarrow \Lambda^b_a e_b^\mu$$

$$\begin{aligned} \gamma^\mu &:= e_a^\mu \gamma^a \longrightarrow \Lambda^b_a e_b^\mu \gamma^a \\ &= u_\Lambda e_b^\mu \gamma^b u_\Lambda^{-1} \end{aligned}$$

Kahler Dirac Fermions

- $dx^\mu \lrcorner dx^\nu = dx^\mu \wedge dx^\nu + g^{\mu\nu}$
- $\gamma^\mu \gamma^\nu = \frac{1}{2} [\gamma^\mu, \gamma^\nu] + g^{\mu\nu}$
- $\phi_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} \rightarrow \phi_{\mu_1 \dots \mu_p} \gamma^{[\mu_1 \dots \mu_p]}$
- $\int * \bar{\Phi} (d - * d *) \Phi \rightarrow \int d^4x \sqrt{g} \text{Tr} [\bar{\Psi} i \gamma^\mu D_\mu \Psi]$

Lattice Action for Kahler Dirac Fermions

$$\bullet S = \int d^4x \sqrt{g} \operatorname{Tr} [\bar{\Psi} i \gamma^\mu \underbrace{(\partial_\mu + [\omega_\mu, -])}_{\downarrow} \Psi]$$

$$\partial_\mu + \overrightarrow{\omega}_\mu - \overleftarrow{A}_\mu$$

$\omega_\mu \rightarrow$ spin connection for spacetime $\operatorname{spin}(4)$

$A_\mu \rightarrow$ gauge connection for $\operatorname{su}(2)_L \times \operatorname{su}(2)_R$

Lattice Action for Kahler Dirac Fermions

- $S = \int d^4x \sqrt{g} \text{Tr} [\bar{\Psi} i \gamma^\mu (\partial_\mu + \vec{W}_\mu - \vec{A}_\mu) \Psi]$

$\Psi \rightarrow$

	$\rightarrow$ taste index	
$\begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$	$\downarrow$ spin index	
$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}$		
$\underbrace{\hspace{10em}}$		
4 copies of Dirac spinor		

All d.o.f. on the
lattice accounted
for!!

Lattice Action for Kahler Dirac Fermions

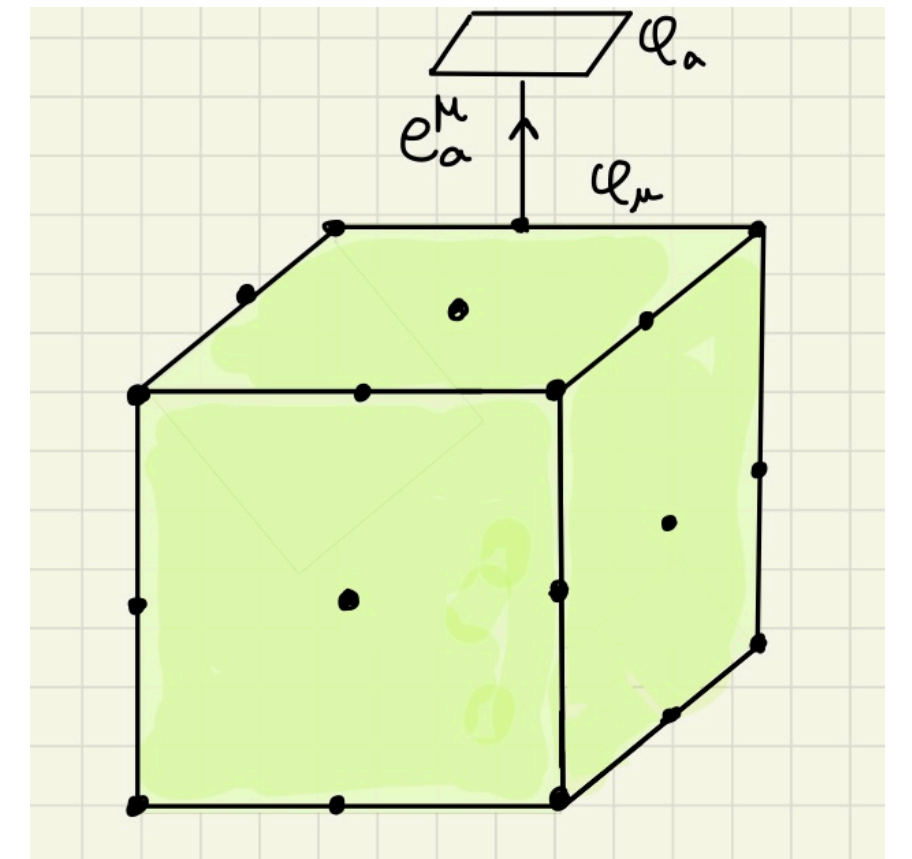
- $$S = \int d^4x \sqrt{g} \text{Tr} [\bar{\Psi} i \gamma^\mu (\partial_\mu + \vec{\omega}_\mu - \vec{A}_\mu) \Psi]$$

↓

- $$\sum_P \sum_{C_P} \text{Tr} \left[\bar{\Psi}(C_P) \left\{ \sum_{C_{P-1}} W(C_{P-1}, C_P) \Psi(C_{P-1}) + \sum_{C_{P+1}} W(C_P, C_{P+1}) \Psi(C_{P+1}) \right\} \right]$$

- Covariant difference operator

$$W(C_{P-1}, C_P) \rightarrow U_\lambda W(C_{P-1}, C_P) U_\lambda^{-1}$$



Cubic Lattice with KD fermions

Chiral Symmetry and the Anomaly

- Chirality operator for polyforms $\Gamma(\Phi_p) = (-1)^p \Phi_p$
 $\Gamma(\Psi) = \gamma_5 \Psi \gamma_5$
- The action is invariant under $\Psi \rightarrow e^{i\alpha\Gamma} \Psi$.
- Reduced KD fermion:

$$S_{\text{RKD}} = \int d^4x \text{Tr} \underbrace{\begin{pmatrix} 0 & O_1 \\ O_2 & 0 \end{pmatrix}}_{\chi_R^{\text{RKD}}} i \not{D} \underbrace{\begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}}_{\Psi_L^{\text{RKD}}};$$

$$E_1 = \Psi_L^i; \quad E_2 = \Psi_R^A$$

$$O_1 = \Psi_L^A; \quad O_2 = \Psi_R^i$$

Chiral Symmetry and the Anomaly

- Chirality operator for polyforms $\Gamma(\phi_p) = (-1)^p \phi_p$
 $\Gamma(\psi) = \gamma_5 \psi \gamma_5$
- The action is invariant under $\psi \rightarrow e^{i\alpha\Gamma} \psi$.
- The extra phase picked by the path-integral $Z = \int D\psi e^{-S_E}$ is the anomaly $\mathcal{A}$.

$$\mathcal{A} = e^{i\alpha(N_0 - N_1 + N_2 - N_3 + N_4)} = e^{i\alpha\chi} \quad \begin{matrix} \nearrow \text{Euler characteristic} \\ \chi \propto \int \text{tr}(R \wedge R) \\ \propto \text{ind}(\mathcal{D}) \end{matrix}$$

Summary

- Chiral fermions come with many issues such as fermion doubling, no-go theorems, and sign problems.
- Kahler Dirac fermions provide a nice discretization that preserves the topological properties of the continuum and is doubler-free.
- The left-right symmetric model or the Pati-Salam model naturally emerges from KD fermions and can be realized as chiral gauge theories on the lattice.
- The field theoretic action for KD fermions still suffers through a sign problem which is overcome through a natural generalization to the worldline KD action.

Sign Problem and the Worldline Action

- Worldline for Dirac fermions

$$S = \int d\tau \frac{1}{2} \left(\frac{1}{e} \dot{x}_\mu \dot{x}^\mu - i \psi_\mu \dot{\psi}^\mu - \frac{i\chi}{e} \psi^\mu \dot{x}_\mu \right)$$

- Equations ($\chi = 0$):
 $\ddot{x}^\mu = 0$; $\dot{\psi}^\mu = 0$

Constraints ($\chi = 0$)

$$\psi^\mu \dot{x}_\mu = 0 \quad ; \quad \dot{x}_\mu \dot{x}^\mu = 0$$

- Quantization

$$\{\psi_\mu, \psi_\nu\} = 2\eta_{\mu\nu} \quad \Rightarrow \quad \psi_\mu \longleftrightarrow \gamma_\mu$$

Sign Problem and the Worldline Action

- Worldline for Kahler-Dirac fermions

$$S = \int d\tau \frac{1}{2} \left(\frac{1}{e} \dot{x}_\mu \dot{x}^\mu - i \psi_{\mu_1 \dots \mu_p} \dot{\psi}^{\mu_1 \dots \mu_p} - \frac{i\chi}{e} \psi^\mu \dot{x}_\mu \right)$$

- Equations ($\chi = 0$):
 $\ddot{x}^\mu = 0$; $\dot{\psi}^{\mu_1 \dots \mu_p} = 0$

Constraints ($\chi = 0$)

$$\psi^\mu \dot{x}_\mu = 0 ; \quad \dot{x}_\mu \dot{x}^\mu = 0$$

- Quantization

$$\{\psi^{\mu_1 \dots \mu_p}, \psi_{\nu_1 \dots \nu_p}\} = \delta_p^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_p} \Rightarrow \psi^{\mu_1 \dots \mu_p} \leftrightarrow \gamma^{[\mu_1 \dots \mu_p]}$$

Sign Problem and the Worldline Action

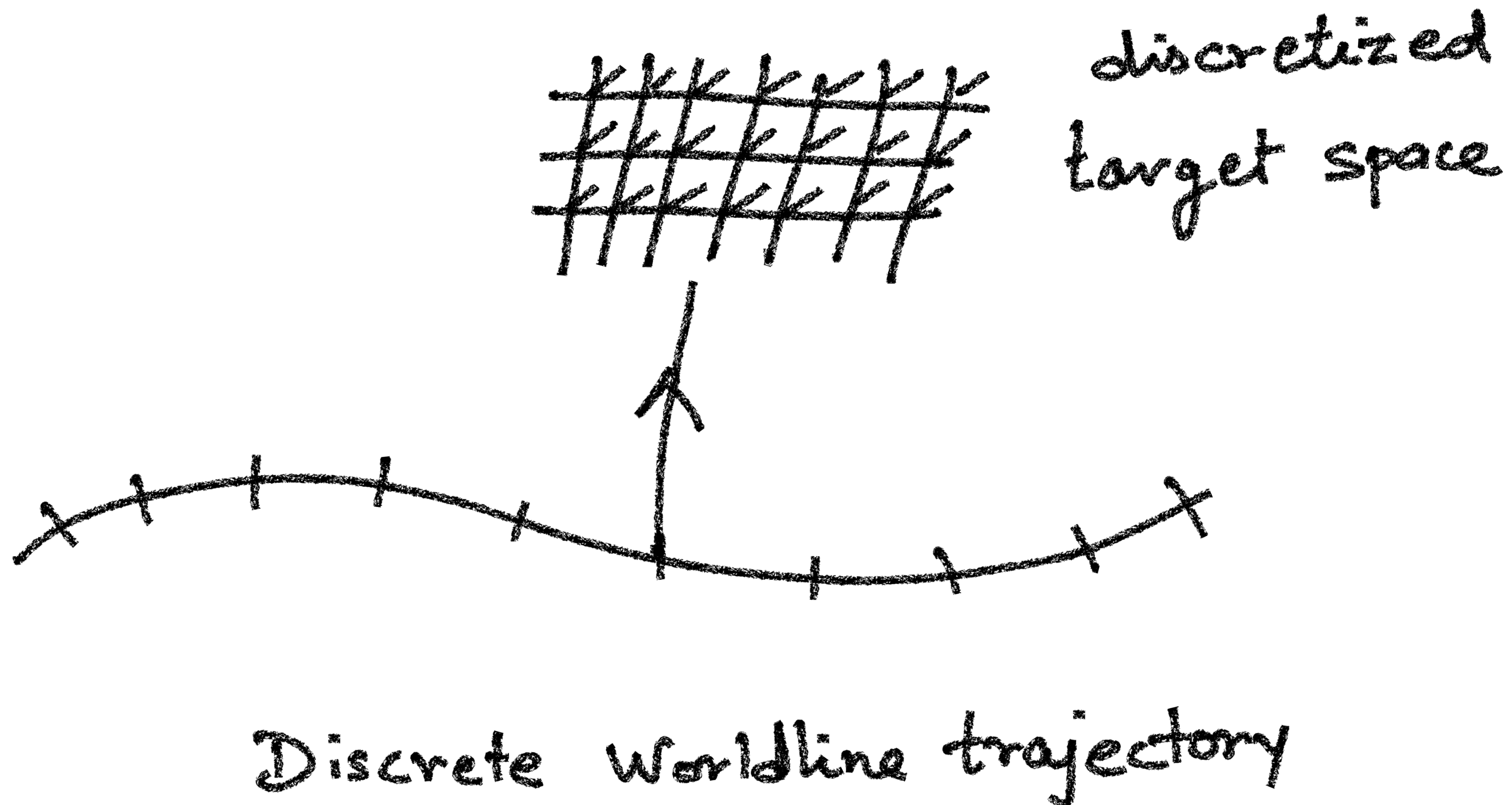
- Worldline for reduced Kahler-Dirac fermions,

$$S = \int d\tau \frac{1}{2} \left(\frac{1}{e} \dot{x}_\mu \dot{x}^\mu - i \psi_{\mu_1 \dots \mu_p} \dot{\psi}^{\mu_1 \dots \mu_p} - \frac{i\chi}{e} \psi^\mu \dot{x}_\mu + (1 + (-1)^p) \lambda_{\mu_1 \dots \mu_p} \psi^{\mu_1 \dots \mu_p} \right)$$

- A simple Lagrange multiplier sets half of the fermions to zero and makes the action chiral.
- Unlike the field theoretic construction, the worldline chiral action does not have a sign problem in the Euclidean signature.
- This happens because the worldline formulation is a theory of Grassmann valued vectors instead of spinors. Just like KD fermions are Grassmann valued polyforms.
- Wick rotation is simply given in terms of rotation the time direction in the metric of the target space.

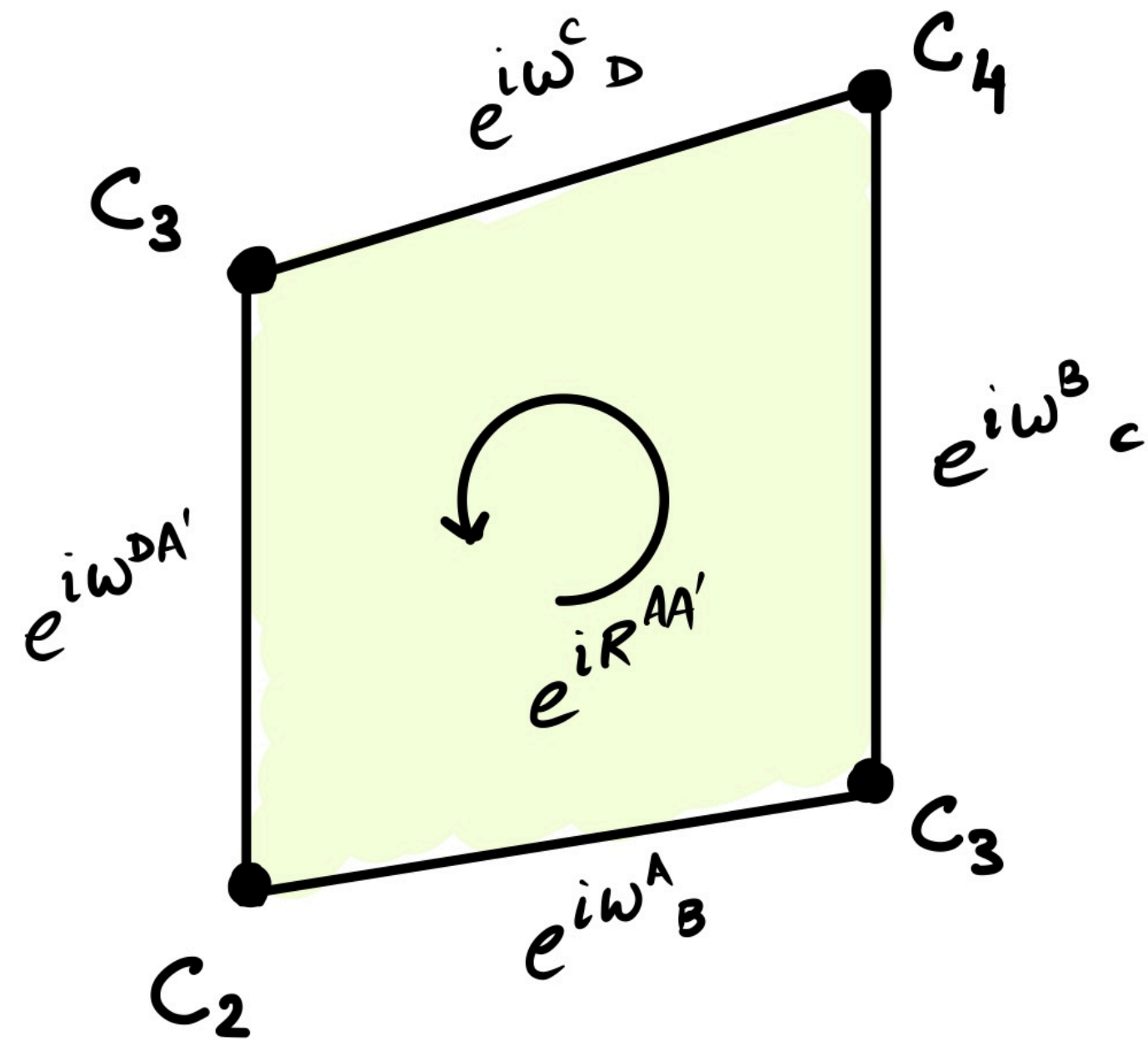
Worldline discretization

- After usual discretization of the target space, one could further discretize the worldline time leading to an action with random-walks and 1D Ising model.



Thank You!

Curvature two-form on the lattice



Curvature 2-cochain

- $\exp(R^{AA'}) = \exp(\oint \omega)$



- $\exp(R^{AA'}) = \prod \exp(\omega^A_B)$



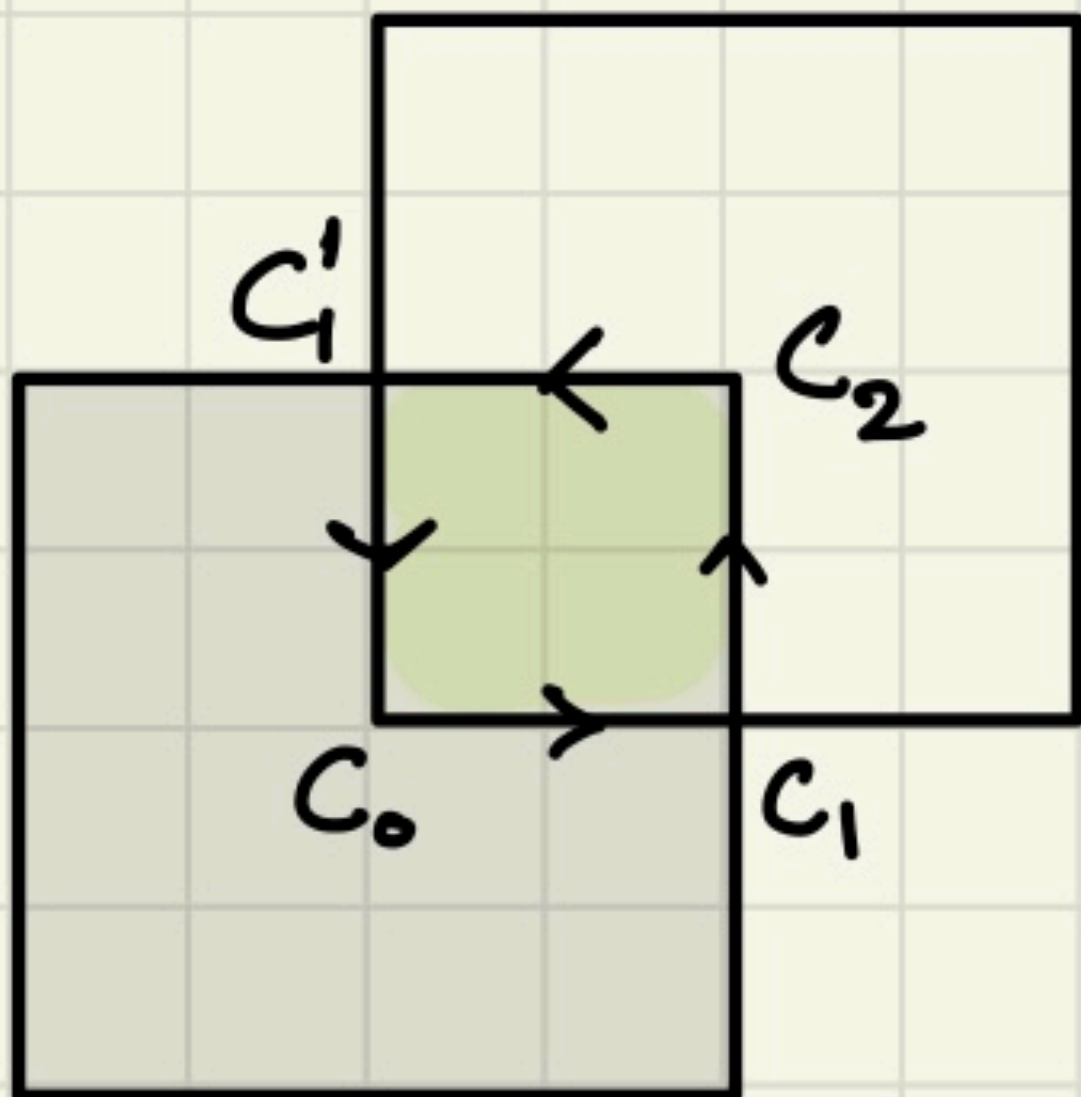
2-cochain



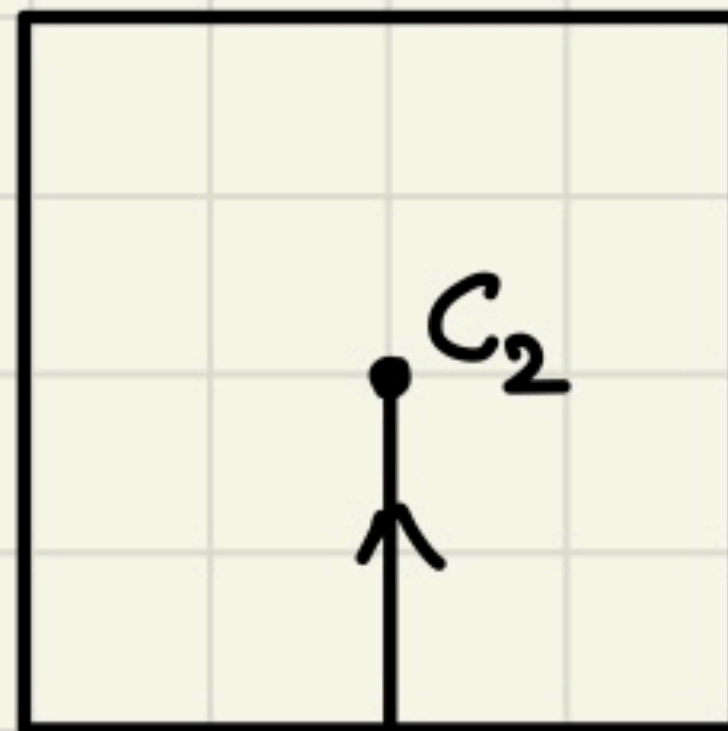
1 cochain

- $R^{AA'}$ is defined on the Plaquette $*C_{D-2} \cap C_D$

$D = 2$



$* C_{D-2} \cap C_D$



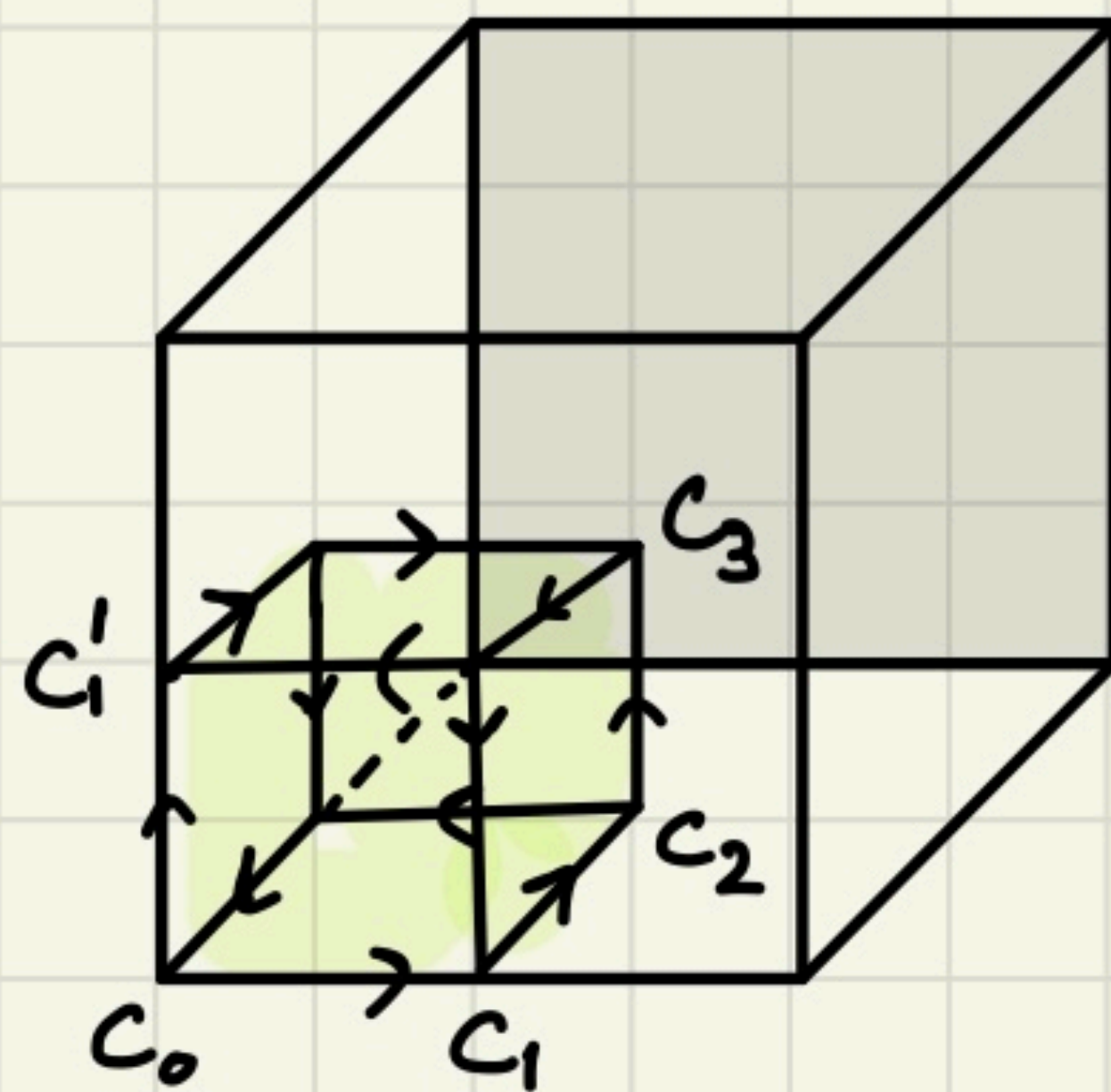
$* C_{D-1} \cap C_D$



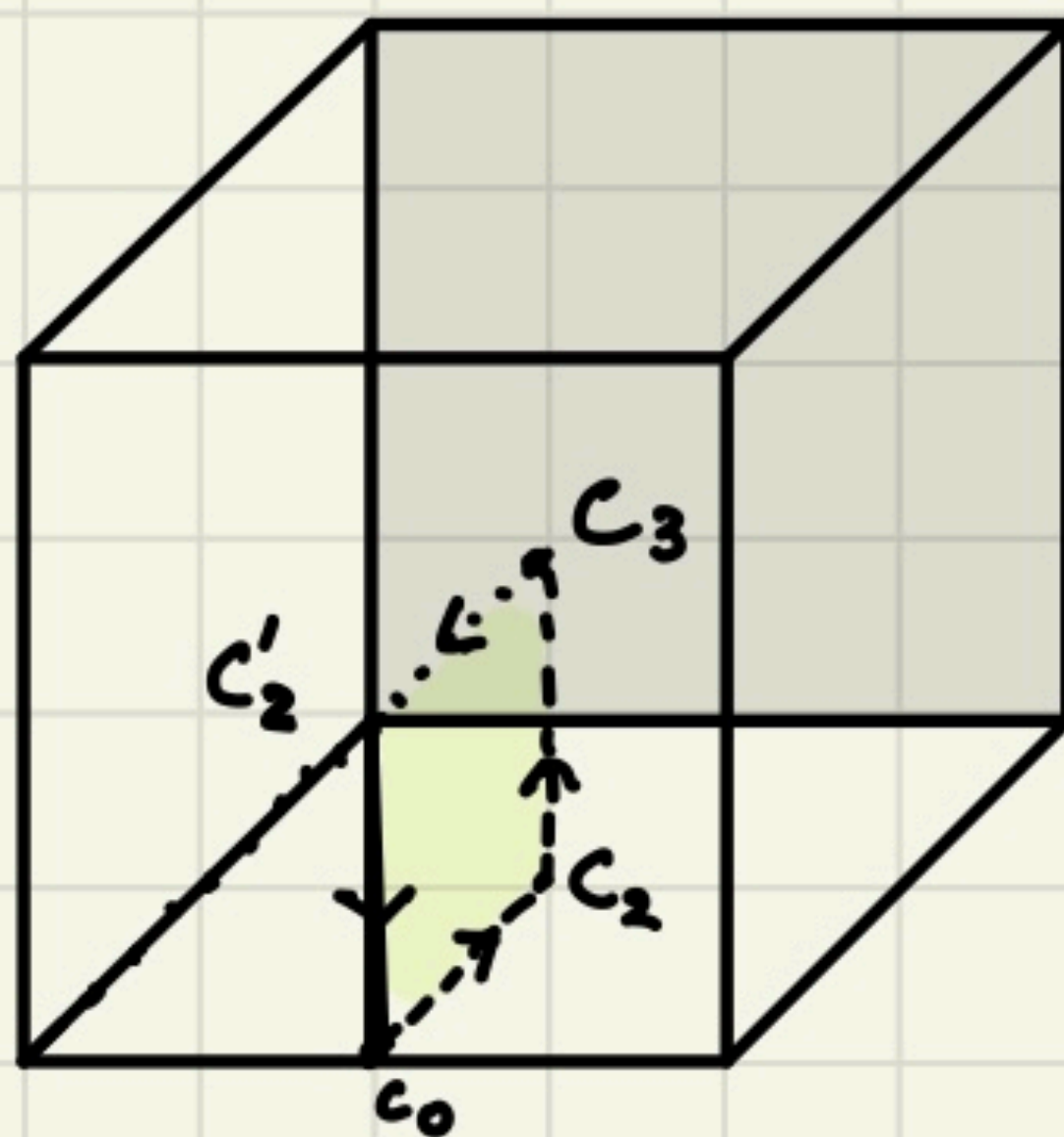
$* C_D \cap C_D$

Different intersections in 2 dimensions

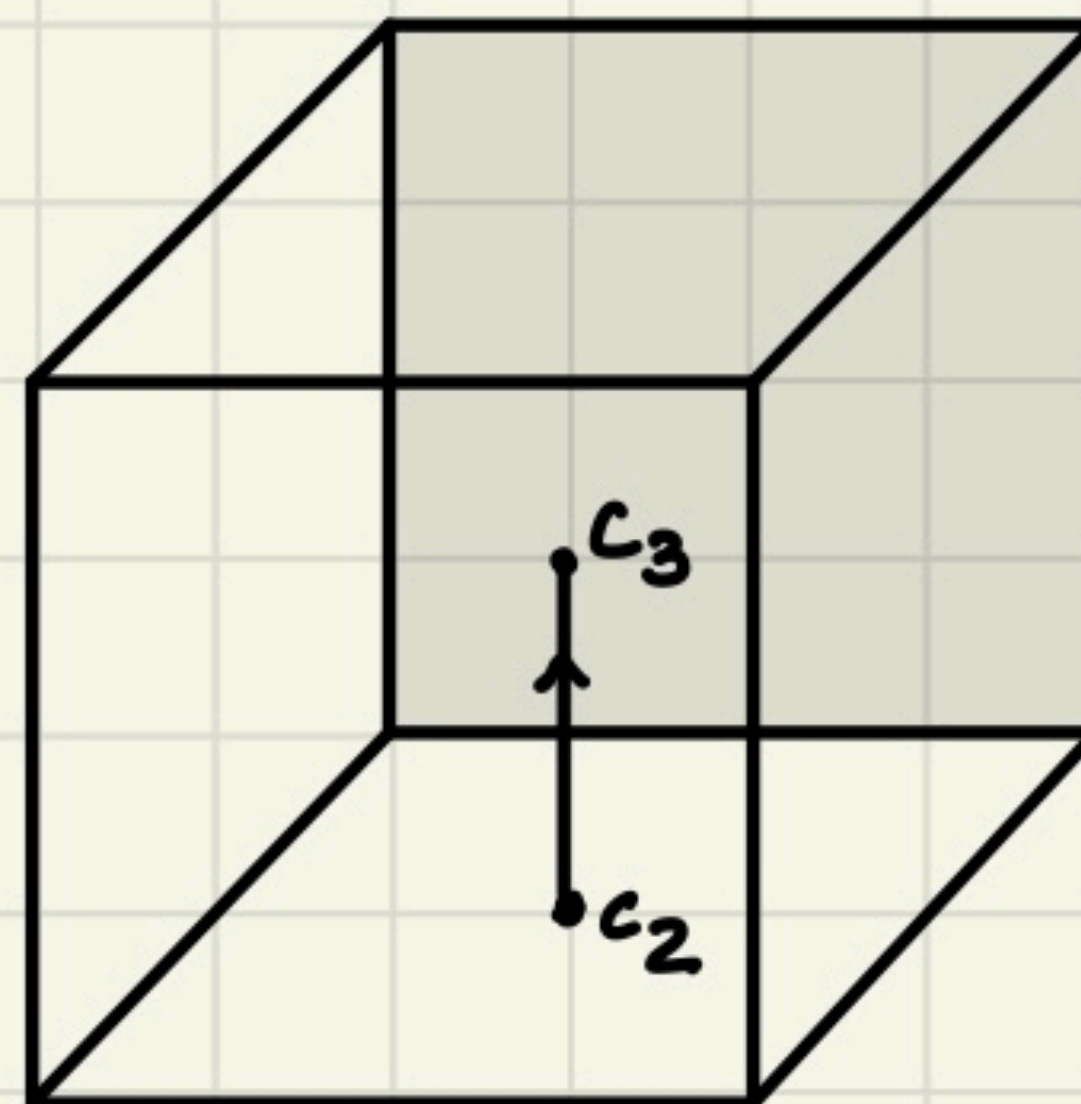
$$D=3$$



$$* C_{D-3} \cap C_D$$



$$* C_{D-2} \cap C_D$$



$$* C_{D-1} \cap C_D$$

Different intersections in 3 dimensions

Einstein-Hilbert action

- Curvature 2-form $R^{A_1 A_2}$ defined on the plaquette
* $C_{D-2} \cap C_D$
- The "area" form $e^{A_3} \wedge \dots \wedge e^{A_D}$ lives on C_{D-2} .
- The Einstein Hilbert action is

$$S = \sum \exp(R^{A_1 A_2}) \wedge e^{A_3} \wedge \dots \wedge e^{A_D} \in_{A_1, \dots, A_D}$$